Linear Operators and Their Characteristic Representations

Structural Properties of Linear Operators and Matrix Representations

Tushar Kumar Aich

August 7, 2026

Abstract

This expository paper presents a rigorous development of linear operator diagonalizability in finite-dimensional vector spaces. We examine the necessary and sufficient conditions under which an operator admits a diagonal matrix representation relative to an ordered basis. We study the characteristic polynomial, its algebraic properties, and basis-independence across similar matrices. We then study the structure of abstract operators with concrete matrix computations, proving key results regarding eigenvalues, trace and determinant.

1 Introduction

We understand that computations involving diagonal matrices are simple. For a given linear operator T on a finite-dimensional vector space V, we seek answers to the following questions:

1. Does there exist an ordered basis $\mathscr{B}$ for V such that $[T]_{\mathscr{B}}$ is a diagonal matrix?

2. If such a basis exists, how can it be found?

An answer to question 1 leads us to how operator T acts on V, and an answer to question 2 helps us to obtain easy solutions to many problems.

2 Eigenvalues and Eigenvectors

Definition 2.1. A linear operator T on a finite-dimensional vector space V is called **diagonalizable** if there is an ordered basis $\mathscr{B}$ for V such that $[T]_{\mathscr{B}}$ is a diagonal matrix. A square matrix A is called diagonalizable if L_A is diagonalizable.

Here if $\mathscr{B} = \{v_1, v_2, \ldots, v_n\}$ is an ordered basis for V such that $[T]_{\mathscr{B}}$ is a diagonal matrix. Then if $D = [T]_{\mathscr{B}}$ is a diagonal matrix, then for each vector $v_j \in \mathscr{B}$,

$$T(v_j) = \sum_{i=1}^{n} D_{ij} v_i = D_{jj} v_j = \lambda_j v_j$$

where $\lambda_j = D_{jj}$.

Conversely, if $\mathscr{B} = \{v_1, v_2, \ldots, v_n\}$ is an ordered basis for V such that $T(v_j) = \lambda_j v_j$ for some scalars $\lambda_1, \lambda_2, \ldots, \lambda_n$, then

$$[T]_{\mathscr{B}} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

Definition 2.2. Let T be a linear operator on a vector space V. A non-zero vector $v \in V$ is called an **eigenvector** of T if there exists a scalar λ such that $T(v) = \lambda v$. The scalar λ is called the **eigenvalue** of T corresponding to the eigenvector v.

Let $A \in M_{n\times n}(F)$. A non-zero vector $v \in F^n$ is called an eigenvector of A if v is an eigenvector of L_A; that is, if $Av = \lambda v$ for some scalar λ. The scalar λ is called the eigenvalue of A corresponding to the eigenvector v.

Remark. Eigenvectors are also called characteristic vectors and proper vectors, and eigenvalues are also called characteristic values and proper values.

Thus, a vector is an eigenvector of A if and only if it is an eigenvector of L_A. Similarly, a scalar λ is an eigenvalue of A if and only if it is an eigenvalue of L_A.

From the above discussion, we conclude that,

Theorem 2.1. *A linear operator T on a finite-dimensional vector space V is diagonalizable if and only if there exists an ordered basis $\mathscr{B}$ for V consisting of eigenvectors of T. Furthermore, if T is diagonalizable, $\mathscr{B} = \{v_1, v_2, \ldots, v_n\}$ is an ordered basis of eigenvectors of T and $D = [T]_{\mathscr{B}}$, then D is a diagonal matrix and D_{jj} is the eigenvalue corresponding to v_j for $1 \leq j \leq n$.*

Theorem 2.2. *Let $A \in M_{n\times n}(F)$. Then a scalar λ is an eigenvalue of A if and only if* $\det(A-\lambda I_n) = 0$.

Proof. A scalar λ is an eigenvalue of A if and only if there exists a non-zero vector $v \in F^n$ such that v is an eigenvector for A, which implies that $Av = \lambda v$. Now,

$$Av = \lambda v = \lambda(I_n v) \implies (A - \lambda I_n)(v) = 0.$$

Now let $T : F^n \to F^n$ be a transformation defined by left-multiplying any vector v by the matrix $(A - \lambda I_n)$. Thus,

$$T(v) = (A - \lambda I_n)v.$$

Thus we get $T(v) = 0$. Since v is a non-zero vector, $N(T) \neq \{0\}$ and hence T is not one-to-one, which means T is not invertible.

Since T is not invertible, then $\operatorname{rank}(T) < n$. Let $\mathscr{B}$ be any ordered basis for F^n. Then $[T]_{\mathscr{B}} = A-\lambda I_n$. We know $\operatorname{rank}(T) = \operatorname{rank}([T]_{\mathscr{B}})$, thus

$$\operatorname{rank}(A - \lambda I_n) = \operatorname{rank}([T]_{\mathscr{B}}) = \operatorname{rank}(T) < n.$$

We know $\det(A) = 0$ if $\operatorname{rank}(A) < n$, thus

$$\det(A - \lambda I_n) = 0.$$

$\square$

Definition 2.3. Let $A \in M_{n\times n}(F)$. The polynomial $f(t) = \det(A - tI_n)$ is called the **characteristic polynomial** of A.

Now if we consider two similar matrices A and B such that $B = P^{-1}AP$ for some invertible matrix P. Then,

$$\begin{aligned}
\det(B - \lambda I_n) &= \det(P^{-1}AP - \lambda I_n) \\
&= \det\big(P^{-1}AP - \lambda(P^{-1}I_nP)\big) \\
&= \det\big(P^{-1}AP - P^{-1}(\lambda I_n)P\big) \\
&= \det\big(P^{-1}(A - \lambda I_n)P\big) \\
&= \det(P^{-1}) \cdot \det(A - \lambda I_n) \cdot \det(P) \\
&= \frac{1}{\det(P)} \cdot \det(P) \cdot \det(A - \lambda I_n) \\
&= \det(A - \lambda I_n).
\end{aligned}$$

Thus, similar matrices have the same characteristic polynomial.

Now let $T : V \to V$ be linear, and let $\mathscr{B}$ and $\mathscr{C}$ be ordered bases for V, and let $[T]_{\mathscr{B}} = A$ and $[T]_{\mathscr{C}} = B$. Then for some invertible matrix Q, by change of basis theorem for linear transformations,

$$[T]_{\mathscr{C}} = Q^{-1}[T]_{\mathscr{B}}Q \quad \text{or} \quad B = Q^{-1}AQ.$$

Since above we showed that similar matrices have the same characteristic polynomial, the characteristic polynomial of a linear operator on a finite-dimensional vector space is independent of the choice of basis. This helps us define the characteristic polynomial as:

Definition 2.4. Let T be a linear operator on an n-dimensional vector space V with ordered basis $\mathscr{B}$. We define the **characteristic polynomial** $f(t)$ of T to be the characteristic polynomial of $A = [T]_{\mathscr{B}}$. That is,

$$f(t) = \det(A - tI_n).$$

Remark. Here, if T is a linear operator on V and $\mathscr{B}$ is an ordered basis for V then λ is an eigenvalue for V if and only if λ is an eigenvalue for $[T]_{\mathscr{B}}$. We denote the characteristic polynomial of an operator T by $\det(T - \lambda I_n)$.

Theorem 2.3. *Let $A \in M_{n\times n}(F)$.*

1. *The characteristic polynomial of A is a polynomial of degree n with leading coefficient $(-1)^n$.*

2. *A has at most n distinct eigenvalues.*

Proof. 1. Let $A \in M_{1\times 1}(F)$ where $A = [A_{11}]$. Then the characteristic polynomial is,

$$f(t) = \det(A - tI_1) = A_{11} - t = (-1)^1 t^1 + A_{11}$$

which is a polynomial of degree 1 with leading coefficient $(-1)^1$. Thus the theorem holds true for $n = 1$. Now let us assume that the theorem holds true for all $(n-1) \times (n-1)$ matrices over F. Now let $A \in M_{n \times n}(F)$ where $n \geq 2$, and let $B = A - tI_n$. So the entries of B are given by,

$$B_{ij} = \begin{cases} A_{ij} - t, & i = j \\ A_{ij}, & i \neq j \end{cases}$$

Now let us compute the characteristic polynomial for A by $A - tI_n$, which is also B. Thus,

$$\begin{aligned} f(t) = \det(B) &= \sum_{j=1}^{n} (-1)^{1+j} B_{1j} \det(\widetilde{B}_{1j}) \\ &= (-1)^{1+1} B_{11} \det(\widetilde{B}_{11}) + \sum_{j=2}^{n} (-1)^{1+j} B_{1j} \det(\widetilde{B}_{1j}) \\ &= (A_{11} - t) \det(\widetilde{B}_{11}) + \sum_{j=2}^{n} (-1)^{1+j} B_{1j} \det(\widetilde{B}_{1j}). \end{aligned}$$

Now $\widetilde{B}_{11}$ is obtained by deleting the 1st row and 1st column of $A - tI_n$, which corresponds to the characteristic matrix of an $(n-1) \times (n-1)$ submatrix of A. By inductive hypothesis,

$$\det(\widetilde{B}_{11}) = (-1)^{n-1} t^{n-1} + g(t)$$

where $g(t)$ is a polynomial such that $\deg(g) \leq n - 2$. Now for $j \geq 2$, the entries $B_{1j} = A_{1j}$ which are independent of t, and the submatrices $\widetilde{B}_{1j}$ lose two diagonal entries of 1 and j, so there are at most $n - 2$ diagonal entries. Thus, the degree of $\det(\widetilde{B}_{1j}) \leq n - 2$. Thus we can write $\sum_{j=2}^{n} (-1)^{1+j} B_{1j} \det(\widetilde{B}_{1j})$ as a polynomial $q(t)$ of degree at most $n - 2$.
Thus, $\deg(q(t)) \leq n - 2$. Therefore,

$$\begin{aligned} \det(B) &= (A_{11} - t)\left((-1)^{n-1} t^{n-1} + g(t)\right) + q(t) \\ &= (-1)^{n-1} A_{11} t^{n-1} + A_{11} g(t) + (-1)^n t^n - t g(t) + q(t). \end{aligned}$$

Here every term except $(-1)^n t^n$ is a polynomial of degree at most $n - 1$. Thus,

$$\det(B) = (-1)^n t^n + r(t)$$

where $\deg(r(t)) \leq n - 1$. Thus $f(t)$ is a polynomial of degree n with leading coefficient $(-1)^n$. Hence the theorem holds true for all $n \times n$ matrices.

2. If we let λ as some scalar such that λ is an eigenvalue for A, then by Theorem 2.2,

$$\det(A - \lambda I_n) = 0.$$

Thus we can say that the eigenvalues are roots of the characteristic polynomial. Since it is a polynomial of degree n and the eigenvalue of A corresponds to a distinct root of the polynomial. Since we know a polynomial of degree n has at most n distinct roots. Thus A has at most n distinct eigenvalues.

□

Remark. We can denote the characteristic polynomial $f(t)$ of a linear operator by

$$f(t) = (-1)^n t^n + a_{n-1}t^{n-1} + \ldots + a_1 t + a_0$$

Theorem 2.4. *For any square matrix A, A and A^T have the same characteristic polynomial.*

Proof. Let $A \in M_{n\times n}(F)$ and let us take some non-zero scalar t. Then,

$$(A - tI_n)^T = A^T - (tI_n)^T = A^T - t(I_n)^T = A^T - tI_n.$$

Thus,

$$\begin{aligned} &\det\left((A - tI_n)^T\right) = \det(A^T - tI_n) \\ \implies &\det(A - tI_n) = \det(A^T - tI_n) \quad [\because \det(A) = \det(A^T)] \end{aligned}$$

Hence, A and A^T have the same characteristic polynomial and hence will have same eigenvalues. □

Theorem 2.5. *Let T be a linear operator on a vector space V and let x be an eigenvector of T corresponding to the eigenvalue λ. For any positive integer m, then x is an eigenvector of T^m corresponding to the eigenvalue λ^m.*

Proof. We have x is an eigenvector of T corresponding to λ. Then $x \neq 0$ and $T(x) = \lambda x$. Applying the transformation again on $T(x)$, we get

$$\begin{aligned} &T(T(x)) = \lambda(T(x)) = \lambda(\lambda x) = \lambda^2 x \\ \implies &T^2(x) = \lambda^2 x. \end{aligned}$$

Thus the theorem holds true for $m = 1, 2$. Let us assume that the theorem holds true for applying the transformation $m - 1$ times. Thus,

$$T^{m-1}(x) = \lambda^{m-1}x$$

Applying the transformation again on $T^{m-1}(x)$, we get

$$\begin{aligned} &T(T^{m-1}(x)) = \lambda(T^{m-1}(x)) = \lambda(\lambda^{m-1}x) = \lambda^m x \\ \implies &T^m(x) = \lambda^m x. \end{aligned}$$

Thus the theorem holds true for applying the transformation m-times for $m \in \mathbb{Z}^+$. □

Corollary 2.5.1. *Let $A \in M_{n\times n}(F)$ and let $x \in F^n$ be an eigenvector of A corresponding to the eigenvalue λ. Then for any positive integer m, x is an eigenvector of A^m corresponding to the eigenvalue λ^m.*

Proof. Let $V = F^n$ and define the left-multiplication operator $L_A : V \to V$ by $L_A(y) = Ay$ for all $y \in V$.

Since x is an eigenvector of A corresponding to λ, we have $x \neq 0$ and $Ax = \lambda x$. Thus,

$$L_A(x) = Ax = \lambda x$$

which implies that x is an eigenvector of L_A corresponding to λ.

By Theorem 2.5,

$$L_A^m(x) = \lambda^m x$$

Since $L_A^m = L_{A^m}$, it follows that:

$$A^m x = L_{A^m}(x) = L_A^m(x) = \lambda^m x$$

Because $x \neq 0$ and $A^m x = \lambda^m x$, x is an eigenvector of A^m corresponding to the eigenvalue λ^m. □

Theorem 2.6. *Let A be an $n \times n$ matrix with characteristic polynomial*

$$f(t) = (-1)^n t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0.$$

Then $f(0) = a_0 = \det(A)$. Also A is invertible if and only if $a_0 \neq 0$.

Proof. We have

$$\begin{aligned} & f(t) = (-1)^n t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0 \\ \implies & f(0) = (-1)^n 0^n + a_{n-1}0^{n-1} + \cdots + a_1 0 + a_0 \\ \implies & f(0) = a_0 \quad . \end{aligned} \tag{1}$$

We also know the characteristic polynomial of an $n \times n$ matrix A is defined by

$$\begin{aligned} & f(t) = \det(A - tI_n) \\ \implies & f(0) = \det(A - 0 \cdot I_n) \\ \implies & f(0) = \det(A) \quad . \end{aligned} \tag{2}$$

From (1) and (2),

$$f(0) = a_0 = \det(A).$$

We know A is invertible if and only if $\det(A) \neq 0$. Thus A is invertible if and only if $a_0 \neq 0$. □

Theorem 2.7. *Let A and $f(t)$ be as in Theorem 2.6. Then,*

1. *$f(t) = (A_{11} - t)(A_{22} - t) \ldots (A_{nn} - t) + q(t)$, where $q(t)$ is a polynomial of degree at most $n - 2$.*
2. *$\operatorname{tr}(A) = (-1)^{n-1} a_{n-1}$.*

Proof. 1. Let $A \in M_{2\times 2}(F)$ such that

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

Then the characteristic polynomial of A is given by

$$\begin{aligned} f(t) = \det(A - tI_2) &= \det\left(\begin{bmatrix} A_{11} - t & A_{12} \\ A_{21} & A_{22} - t \end{bmatrix}\right) \\ &= (A_{11} - t)(A_{22} - t) - A_{12}A_{21} \\ &= (A_{11} - t)(A_{22} - t) - A_{12}A_{21}t^0 \\ &= (A_{11} - t)(A_{22} - t) + q(t) \end{aligned}$$

where $q(t)$ is a constant or a polynomial of degree at most 0. Thus the theorem is true for 2×2 matrices. Now let us assume that the statement is true for $(n-1) \times (n-1)$ matrices where $q(t)$ is a polynomial of degree at most $(n-1) - 2 = n - 3$.

Now let $A \in M_{n\times n}(F)$, then its characteristic polynomial is given by

$$f(t) = \det(A - tI_n)$$

Let $B = A - tI_n$. Then the entries of B are given by

$$B_{ij} = \begin{cases} A_{ij} - t, & i = j \\ A_{ij}, & i \neq j. \end{cases}$$

Thus,

$$\begin{aligned} \det(B) &= \sum_{j=1}^{n} (-1)^{1+j} B_{1j} \det(\widetilde{B}_{1j}) \\ &= (A_{11} - t)\det(\widetilde{B}_{11}) + \sum_{j=2}^{n} (-1)^{1+j} B_{1j} \det(\widetilde{B}_{1j}). \end{aligned}$$

Here $\widetilde{B}_{11}$ is a matrix obtained by removing 1st row and 1st column from $A - tI_n$ and $\det(\widetilde{B}_{11})$ is the characteristic polynomial of that obtained matrix. By induction hypothesis,

$$\det(\widetilde{B}_{11}) = (A_{22} - t)(A_{33} - t)\ldots(A_{nn} - t) + g(t)$$

where $g(t)$ is a polynomial of at most $n - 3$. Now,

$$\begin{aligned} (A_{11} - t)\det(\widetilde{B}_{11}) &= (A_{11} - t)\left[(A_{22} - t)(A_{33} - t)\ldots(A_{nn} - t) + g(t)\right] \\ &= (A_{11} - t)(A_{22} - t)\ldots(A_{nn} - t) + (A_{11} - t)g(t) \\ &= (A_{11} - t)(A_{22} - t)\ldots(A_{nn} - t) + \left(A_{11}g(t) - tg(t)\right). \end{aligned}$$

Here $r(t) = A_{11}g(t) - tg(t)$ is a polynomial of at most $n - 2$. Therefore,

$$(A_{11} - t)\det(\widetilde{B}_{11}) = (A_{11} - t)(A_{22} - t)\ldots(A_{nn} - t) + r(t).$$

Now for $j \geq 2$, $B_{1j} = A_{1j}$ which are independent of t, and the submatrices $\widetilde{B}_{1j}$ loses 2 of its diagonal entries, so there are at most $n-2$ diagonal entries. Thus $\sum_{j=2}^{n}(-1)^{1+j}B_{1j}\det(\widetilde{B}_{1j})$ is a polynomial, say $p(t)$, of degree at most $n-2$. Hence,

$$\begin{aligned} f(t) &= (A_{11}-t)(A_{22}-t)\dots(A_{nn}-t)+r(t)+p(t) \\ &= (A_{11}-t)(A_{22}-t)\dots(A_{nn}-t)+q(t) \end{aligned}$$

where $q(t)$ is a polynomial of degree at most $n-2$. Thus the theorem holds true for $n\times n$ matrices for $n\in\mathbb{Z}^{+}$.

2. Let $A\in M_{2\times 2}(F)$, then from part (1),

$$\begin{aligned} f(t) &= (A_{11}-t)(A_{22}-t)+q(t) \quad \text{where } \deg(q(t))\leq 2-2=0 \\ &= t^2-(A_{11}+A_{22})t+A_{11}A_{22}+q(t) \\ &= t^2+(-1)^{2-1}\operatorname{tr}(A)t^{2-1}+r(t) \quad [r(t)=A_{11}A_{22}+q(t)]\,. \end{aligned}$$

Thus we observe that for 2×2 matrices its trace appears at t^{2-1} with $(-1)^{2-1}$ in its characteristic polynomial. Let us make an inductive hypothesis that for $(n-1)\times(n-1)$ matrix the trace of that matrix appears at $t^{(n-1)-1}$ with $(-1)^{(n-1)-1}$ in its respective characteristic polynomial. Now let $A\in M_{n\times n}(F)$, then from part (1),

$$f(t)=(A_{11}-t)(A_{22}-t)\dots(A_{nn}-t)+r(t) \quad \text{where } \deg(r(t))\leq n-2.$$

Since $\deg(r(t))\leq n-2$, we can ignore it completely. Thus,

$$f(t)=(A_{11}-t)(A_{22}-t)\dots(A_{(n-1)(n-1)}-t)(A_{nn}-t).$$

By inductive hypothesis,

$$\begin{aligned} f(t) = \big[&(-1)^{n-1}t^{n-1}+(-1)^{n-2}(A_{11}+A_{22}+\dots+A_{(n-1)(n-1)})t^{n-2} \\ &+(\text{lower degree polynomials})\big]\,(A_{nn}-t). \end{aligned}$$

Here we can again ignore the lower degree polynomials. Thus,

$$\begin{aligned} f(t) &= \left[(-1)^{n-1}t^{n-1}+(-1)^{n-2}(A_{11}+A_{22}+\dots+A_{(n-1)(n-1)})t^{n-2}\right](A_{nn}-t) \\ &= (-1)^{n-1}A_{nn}t^{n-1}+(-1)^{n-2}A_{nn}(A_{11}+A_{22}+\dots+A_{(n-1)(n-1)})t^{n-2}+(-1)^{n}t^{n} \\ &\quad +(-1)^{n-1}(A_{11}+A_{22}+\dots+A_{(n-1)(n-1)})t^{n-1} \\ &= (-1)^{n}t^{n}+(-1)^{n-1}(A_{11}+A_{22}+\dots+A_{nn})t^{n-1}+(\text{lower degree polynomials}) \\ &= (-1)^{n}t^{n}+(-1)^{n-1}\operatorname{tr}(A)t^{n-1}+(\text{lower degree polynomials}). \end{aligned}$$

Thus our hypothesis is true. Now from Theorem 2.6,

$$f(t)=(-1)^{n}t^{n}+a_{n-1}t^{n-1}+(\text{lower degree polynomials}).$$

Upon comparing, we get

$$(-1)^{n-1}\operatorname{tr}(A) = a_{n-1} \implies \operatorname{tr}(A) = (-1)^{n-1}a_{n-1}.$$

□

Remark. The characteristic polynomial for an $n \times n$ matrix can be written as

$$f(t) = (-1)^n t^n + (-1)^{n-1}\operatorname{tr}(A)t^{n-1} + a_{n-2}t^{n-2} + \ldots + a_1 t + \det(A).$$

For n=2, we get,

$$f(t) = (-1)^2 t^2 + (-1)^{2-1} tr(A) t^{2-1} + \det(A) = t^2 - \operatorname{tr}(A)t + \det(A).$$

Theorem 2.8. *Let T be a linear operator on V, and let λ be an eigenvalue of T. A vector $v \in V$ is an eigenvector of T, corresponding to λ if and only if $v \neq 0$ and $v \in N(T - \lambda I)$.*

Proof. Let $v \in V$ be an eigenvector for T. Then by definition of eigenvector $v \neq 0$ and

$$\begin{aligned} T(v) &= \lambda v \\ &= \lambda I(v) \\ \implies T(v) - (\lambda I)(v) &= 0 \\ \implies (T - \lambda I)(v) &= 0. \end{aligned}$$

Thus, $v \in N(T - \lambda I)$. Now let us assume $v \neq 0$ and $v \in N(T - \lambda I)$. Thus,

$$\begin{aligned} (T - \lambda I)(v) &= 0 \\ \implies T(v) - \lambda I(v) &= 0 \\ \implies T(v) - \lambda v &= 0 \\ \implies T(v) &= \lambda v. \end{aligned}$$

Thus by definition $v \in V$ is an eigenvector for T. □

Remark. The set $N(T - \lambda I)$ is called the **eigenspace** of T corresponding to the eigenvalue λ, denoted by E_λ. It consists of the zero vector together with all eigenvectors of T corresponding to λ.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\phi_{\mathscr{B}}"] & V \arrow[d, "\phi_{\mathscr{B}}"] \\ F^n \arrow[r, "L_A"] & F^n \end{tikzcd}$$

Now, to find the eigenvectors of a linear operator T on an n-dimensional vector space, let us select an ordered basis $\mathscr{B}$ for V and let $A = [T]_{\mathscr{B}}$. Here we use a commutative diagram. Here for $\phi_{\mathscr{B}}(v) = [v]_{\mathscr{B}}$

the coordinate vector of v relative to $\mathscr{B}$. We define $v \in V$ is an eigenvector of T corresponding to λ if and only if $\phi_{\mathscr{B}}(v)$ is an eigenvector of A corresponding to λ.

$$A\phi_{\mathscr{B}}(v) = L_A\phi_{\mathscr{B}}(v) = \phi_{\mathscr{B}}T(v) = \phi_{\mathscr{B}}(\lambda v) = \lambda\phi_{\mathscr{B}}(v).$$

Theorem 2.9. *Let T be a linear operator on a finite dimensional vector space V over a field F, let $\mathscr{B}$ be an ordered basis for V and let $A = [T]_{\mathscr{B}}$. In reference to*

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\phi_{\mathscr{B}}"] & V \arrow[d, "\phi_{\mathscr{B}}"] \\ F^n \arrow[r, "L_A"] & F^n \end{tikzcd}$$

1. *If $v \in V$ and $\phi_{\mathscr{B}}(v)$ is an eigenvector of A corresponding to the eigenvalue λ, then v is an eigenvector of T corresponding to λ.*

2. *If λ is an eigenvalue of A (and hence of T), then a vector $y \in F^n$ is an eigenvector of A corresponding to λ if and only if $\phi_{\mathscr{B}}^{-1}(y)$ is an eigenvector of T corresponding to λ.*

Proof. 1. If $\phi_{\mathscr{B}}(v)$ is an eigenvector of A corresponding to λ, then $\phi_{\mathscr{B}}(v) \neq 0$. As $\phi_{\mathscr{B}}$ is an isomorphism, $v \neq 0$ and also,

$$\begin{aligned} A\phi_{\mathscr{B}}(v) &= \lambda\phi_{\mathscr{B}}(v) \\ \implies [T]_{\mathscr{B}}[v]_{\mathscr{B}} &= \lambda[v]_{\mathscr{B}}. \end{aligned}$$

As $\phi_{\mathscr{B}}$ is linear and we know if $U, T \in \mathscr{L}(V)$, then $[UT]_{\mathscr{B}} = [U]_{\mathscr{B}}.[T]_{\mathscr{B}}$. Thus,

$$\begin{aligned} \implies [T(v)]_{\mathscr{B}} &= [\lambda v]_{\mathscr{B}} \\ \implies \phi_{\mathscr{B}}(T(v)) &= \phi_{\mathscr{B}}(\lambda v). \end{aligned}$$

As $\phi_{\mathscr{B}}$ is one-to-one,

$$\implies T(v) = \lambda v.$$

Thus, v is an eigenvector of T corresponding to λ.

From part (1) and theory we conclude that $\phi_{\mathscr{B}}(v)$ is an eigenvector of A corresponding to λ if and only if v is an eigenvector of T corresponding to λ.

2. Since we know $\phi_{\mathscr{B}} : V \to F^n$ is one-to-one and onto, then $\forall y \in F^n$, $\exists v \in V$ such that $\phi_{\mathscr{B}}(v) = y$, and since $\phi_{\mathscr{B}}$ is an isomorphism, hence it is invertible. Thus $v = \phi_{\mathscr{B}}^{-1}(y)$.

Now if y is an eigenvector of A corresponding to λ, then $y \neq 0$ and since $\phi_{\mathscr{B}}$ is an isomorphism, thus $v = \phi_{\mathscr{B}}^{-1}(y) \neq 0$.

Hence, $y = \phi_{\mathscr{B}}(v)$ is an eigenvector of A corresponding to λ if and only if $v = \phi_{\mathscr{B}}^{-1}(y)$ is an eigenvector of T corresponding to λ.

□

Theorem 2.10. *Let T be a linear operator on a finite dimensional vector space V and let $\mathscr{B}$ be an ordered basis for V. Prove that λ is an eigenvalue of T if and only if λ is an eigenvalue of $[T]_{\mathscr{B}}$.*

Proof. Let $A = [T]_{\mathscr{B}} \in M_{n\times n}(F)$ and let $\phi_{\mathscr{B}} : V \to F^n$ be the coordinate isomorphism defined by $\phi_{\mathscr{B}}(v) = [v]_{\mathscr{B}}$. Since $\mathscr{B}$ is an ordered basis, $\phi_{\mathscr{B}}$ is linear, one-to-one, and onto.

($\Longrightarrow$) Suppose λ is an eigenvalue of T. Then there exists a non-zero eigenvector $v \in V$ such that $T(v) = \lambda v$. Since $\phi_{\mathscr{B}}$ is an isomorphism, $v \neq 0$ implies that $\phi_{\mathscr{B}}(v) = [v]_{\mathscr{B}} \neq 0$ in F^n. Taking the coordinate vectors on both sides of $T(v) = \lambda v$, we obtain:

$$[T(v)]_{\mathscr{B}} = [\lambda v]_{\mathscr{B}} \implies [T]_{\mathscr{B}}[v]_{\mathscr{B}} = \lambda[v]_{\mathscr{B}} \implies A\phi_{\mathscr{B}}(v) = \lambda\phi_{\mathscr{B}}(v).$$

Since $\phi_{\mathscr{B}}(v) \neq 0$, $\phi_{\mathscr{B}}(v)$ is an eigenvector of A corresponding to λ. Hence, λ is an eigenvalue of $A = [T]_{\mathscr{B}}$.

($\Longleftarrow$) Conversely, suppose λ is an eigenvalue of $A = [T]_{\mathscr{B}}$. Then there exists a non-zero vector $y \in F^n$ such that $Ay = \lambda y$. Because $\phi_{\mathscr{B}}$ is an isomorphism, there exists a unique non-zero vector $v \in V$ such that $\phi_{\mathscr{B}}(v) = y$ (that is, $v = \phi_{\mathscr{B}}^{-1}(y) \neq 0$). Substituting $[v]_{\mathscr{B}} = y$ into $Ay = \lambda y$ gives:

$$[T]_{\mathscr{B}}[v]_{\mathscr{B}} = \lambda[v]_{\mathscr{B}} \implies [T(v)]_{\mathscr{B}} = [\lambda v]_{\mathscr{B}}.$$

Since $\phi_{\mathscr{B}}$ is one-to-one, $[T(v)]_{\mathscr{B}} = [\lambda v]_{\mathscr{B}}$ implies $T(v) = \lambda v$. Since $v \neq 0$, v is an eigenvector of T corresponding to λ. Hence, λ is an eigenvalue of T.

Therefore, λ is an eigenvalue of T if and only if λ is an eigenvalue of $[T]_{\mathscr{B}}$. □

Geometric Intuition

Let v be an eigenvector of T and λ be the corresponding eigenvalue. We can think of $W = \text{span}(\{v\})$, the 1D subspace of V spanned by v, as a line in V that passes through 0 and v. For any $w \in W$, $w = cv$ for some scalar c. Thus,

$$T(w) = T(cv) = cT(v) = c\lambda v = \lambda w.$$

So T acts on the vectors in W by multiplying each vector by λ.

1. $(\lambda > 1)$: T moves vectors in W farther from 0 by a factor of λ.
2. $(\lambda = 1)$: T acts as identity operator on W.
3. $(0 < \lambda < 1)$: T moves vectors in W closer to 0 by a factor of λ.
4. $(\lambda = 0)$: T acts as 0 transformation on W.
5. $(\lambda < 0)$: T reverses orientation of W, i.e., T moves vectors in W from one side of 0 to other.

3 Conclusion

In this paper, we have detailed the foundational framework governing the diagonalizability of linear operators on finite-dimensional vector spaces. By establishing the equivalence between operator diagonalizabilit and the existence of a basis consisting entirely of eigenvectors, we linked abstract linear transformations directly to structured matrix representations. The characteristic polynomial serves as a central invariant tool, demonstrating that spectral properties, trace, and determinant remain preserved under basis changes and matrix similarities. Furthermore, using coordinate isomorphisms, we connected coordinate vectors in F^n to underlying spaces, showing how algebraic conditions on characteristic roots dictate the geometric scaling and orientation behavior of linear operators.

References

[1] Friedberg, S. H., Insel, A. J., & Spence, L. E. (2018). *Linear Algebra* (5th ed.). Pearson.